

Q1: The 2^n ladder, via analytic combinatorics
 Reply to Gunnar’s note of 29 Apr 2026

0. Framing: what we have actually been claiming

Before answering the specific 2^n question, it is worth restating the claim cleanly, because the messy-diagrammatics complaint comes from a place where two of our claims were not separated. We have been making *three* distinct claims, each one strictly on top of the previous:

1. **Claim 1 (algorithmic, Layer 1 of the tutorial).** For a CRN, the Doi shift gives a closed-form vertex generator $Q_r = k_r [\prod_i (1 + \tilde{\psi}_i)^{l_{ri}} - \prod_i (1 + \tilde{\psi}_i)^{k_{ri}}] \prod_i \psi_i^{k_{ri}}$, and from that an algorithm enumerates all amputated vertices and all N -loop topologies. This is just code: vertex dictionary, leg-count bookkeeping, isomorphism-canonicalised graph enumeration. No human-counting of contractions. **The closed-form generator is the input the algorithm consumes; it is not yet AC.**
2. **Claim 2 (AC contribution, Layer 2).** The same closed-form generator, fed through Lagrange inversion / Connes–Kreimer Hopf antipode, gives *symmetry factors and double poles in closed form* — not by drawing each graph, but as residues / antipode evaluations. This is what AC actually adds beyond Layer 1: $|\text{Aut}(\Gamma)|^{-1}$ comes out of Lagrange-inversion overcount on identical subtrees, and the 2-loop Z -factors come out of one rational antipode formula.
3. **Claim 3 (curiosity, Layer 4).** The QFT relevance filters (1PI restriction, external-leg sector, power counting) can in principle be folded into the generating function itself, so the GF *subtracts out* sectors we do not care about rather than us pruning by hand. We have only sketched this — the Legendre transform / multivariate bidegree story — in `legendre_tutorial.tex`. We have not run it end-to-end; for 2-loop hand-curated theories it does not save labour. We mention it because it is where the natural answer to “can the *generating function* (rather than us) decide what is relevant?” sits.

The 2^n ladder is a Claim-2 question (algebra + AC reading-off); the φ^4 Wick-weight question in your §1 is a Claim-1+2 question (algorithm enumerates, AC reads symmetry factor); the philosophical $4 \cdot 3$ vs. $\binom{4}{2}^2 \cdot 2 \cdot 4!/2$ point is the same machine evaluated at different orders of the same generator.

The next three sections (§1–3) walk Claims 1–3 with concrete BRW examples to show what each layer actually delivers. §4 then directly answers the 2^n question. Verification scripts are in `verify_2x2.py` (the H -vertex collapse) and `poc_brw.py` (the BRW vertex/loop enumeration with thesis cross-checks).

1. Claim 1 in action: BRW vertex dictionary and 1-loop bubbles

The Doi shift on the BRW reactions (thesis Eqs. 3.16 + 3.21) collapses to a closed-form vertex generator whose monomial expansion is the seven-vertex dictionary

$$\{V_{\text{branch}}, V_{QC1}, V_{QC2}, V_{QC3}, V_{QC4}, V_{QC5}, V_{QC6}\}$$

with leg counts and signs recorded once and for all. `poc_brw.py` builds this list directly from the stoichiometries and then enumerates all 1-loop bubbles by walking pairs of vertices and pairs of internal lines. No graphs drawn by hand. The output is:

- **36 species-decorated 1-loop bubble topologies**, grouped into **20 external-leg sectors** (e.g. $(0, 0, 2, 2)$ has 1 bubble, $(0, 1, 2, 1)$ has 5, etc.);
- Three are 2-point ($E=2$); seven are 3-point ($E=3$); 27 are distinct (vertex-pair, internal-species-set) skeletons. The “7 distinct 1-loop integrals” on thesis p.91–92 are the 3-point sector exactly.

This is purely algorithmic: closed-form generator $\rightarrow$ vertex dictionary $\rightarrow$ enumerate. It would extend mechanically to 2-loop (more vertex pairs, more internal-line patterns). The point is: *this layer does not need AC at all*. AC’s job starts with the next claim.

2. Claim 2 in action: AC reads off symmetry factors and prunes

Symmetry factors — Lagrange-inversion side. For Reggeon DP, the generator collapses to $G(z) = z(1 + G^2)$. Lagrange inversion gives the entire tree-topology inventory

$$[z^3]G = 1, \quad [z^5]G = 2, \quad [z^7]G = 5,$$

and these counts *already include the* $|\text{Aut}(\Gamma)|^{-1}$ factor per topology — the inversion overcount on identical subtrees is exactly the symmetry weight in the Feynman rule. Total $1+2+5 = 8$ topologies through 2-loop vertex order, with no graphs drawn.

Symmetry factors — BRW side, automorphism counting. For the multi-species BRW the symmetry factor is computed from the bubble’s automorphism group (line-swap if same species/direction; vertex-swap if same type and the line set is symmetric). Concrete output of `poc_brw.py` on the pure-A skeleton (corresponds to the Reggeon-DP 1-loop bubble after quenching the carrying-capacity field):

v_1	v_2	external residue	$s(\Gamma)$
V_{branch}	V_{branch}	$2A_{\text{out}}$	2
V_{branch}	V_{QC4}	$1A_{\text{out}}, 2B_{\text{in}}, 1B_{\text{out}}$	1
V_{QC4}	V_{QC4}	$4B_{\text{in}}, 2B_{\text{out}}$	2
V_{QC5}	V_{QC5}	$2B_{\text{in}}, 2B_{\text{out}}$	2

The boxed $s = 2$ for the pure-branch bubble is thesis Eq. (3.25). The algorithm reproduces it without us touching the diagram.

Closed-form double poles — Hopf antipode. For the same Reggeon-DP topology inventory, the Connes–Kreimer antipode evaluation is one rational formula,

$$Z_X^{(2,2)} = \frac{1}{2} a_X^{(1)} (\beta_1 + a_X^{(1)}), \quad \beta_1 = a_u^{(1)} - 2a_\psi^{(1)} = \frac{3}{2},$$

with the 1-loop residues $a^{(1)} = \{1/4, 1/8, 1/2, 2\}$ giving $\{Z_\psi, Z_\lambda, Z_\tau, Z_u\}^{(2,2)} = \{7/32, 13/128, 1/2, 7/2\}$, matching JT05 Eq. (57). This is the closed-form payoff of Claim 2: not just *counts and symmetry factors*, but the full 2-loop double-pole table from one polynomial. See Q2 for the JT05 cross-check end-to-end.

Where Claim 2 stops. What Layer 2 does *not* give: it does not, by itself, know which graphs are 1PI vs. reducible, which external-leg sector you want, or which power-counting sectors are UV-divergent. Those are QFT-side filters that we currently apply by hand on top of the AC counts. E.g. for the 2-loop vertex sector we know “ $[z^7]G = 5 = 3 \text{ 1PI} + 2 \text{ reducible}$ ” — but we know that from drawing the graphs, not from the generating function.

3. Claim 3 (curiosity): inject the QFT filters into the GF itself

The natural follow-up is whether the *generating function* can be made to subtract out the irrelevant sectors itself, so the algorithmic and AC layers do not need a hand-curated topology table. Three places this could go:

- **1PI via Legendre transform.** Standard QFT: $W[J] = \log Z[J]$ is the connected-graph generator; its Legendre transform $\Gamma[\Phi]$ is the 1PI generator. On the bivariate $\mathcal{Z}(g, r; J, \tilde{J})$, the Legendre transform is a polynomial substitution on the source markers — in principle this gives the 1PI subset directly without graph-by-graph sieving.
- **External-leg sector via bidegree.** Marking sources with $J^a \tilde{J}^b$ tags external legs; reading off $[J^a \tilde{J}^b]$ of Γ projects onto the (a, b) sector. This is the same bidegree-projection trick we use on the H -vertex below.
- **UV relevance via a polynomial inequality on monomial degrees.** Power counting $\omega = dL - 2I + n_\partial$ becomes a linear inequality on the (L, I, n_∂) multidegree of each monomial in the generator; a relevance projector kills the $\omega < 0$ monomials.

`legendre_tutorial.tex` starts to flesh this out: it walks the four layers, with the Legendre / bidegree / power-counting projectors as Layer 4. We have *not* yet run a full theory through Layer 4 end-to-end; the existing 2-loop pipeline applies these filters by hand on top of the AC counts (Q2 §3 documents this honestly). We note the curiosity because it is the principled answer to “why should the GF not handle the filtering itself?” — but we are not yet claiming a worked-out machine.

4. Direct answer to your 2^n question

With the framing of §0–3 in place, your 2^n ladder is a Claim-2 question on the H -vertex generator. You posed the four-term coupling

$$\left[\begin{array}{c} m_1 n_1 \\ \text{Y} \\ m_2 n_2 \end{array} \right] = \delta_{m_2, n_2} \delta_{m_1-1, n_1+1} + \delta_{m_2, n_2} n_1 \delta_{m_1-1, n_1-1} - \delta_{m_2, n_2+1} \delta_{m_1-1, n_1+1} - \delta_{m_2, n_2-1} n_2 \delta_{m_1-1, n_1}$$

with chain boundaries fixed at $(1, 0)$ on both ends and computed (in Mathematica, through three loops) the n -rung sum

$$\sum_{\ell_1, \dots, \ell_{2n}} \begin{array}{c} 1 \ell_1 \\ \text{Y} \\ 0 \ell_2 \end{array} \cdot \left(\begin{array}{c} \ell_1 \ell_3 \\ \text{Y} \\ \ell_2 \ell_4 \end{array} + \begin{array}{c} \ell_2 \ell_4 \\ \text{Y} \\ \ell_1 \ell_3 \end{array} \right) \cdots = 2^n,$$

and asked: *can analytic combinatorics see this 2^n structurally, without an infinite matrix and without “a generating function in four variables — a mess”?*

The AC move. Encode each leg occupation by a marker variable: z_i for input rail i , w_j for output rail j . The vertex coupling becomes the four-variable polynomial

$$\hat{V}(z_1, z_2; w_1, w_2) = \sum_{m_1, m_2, n_1, n_2 \geq 0} V(m_1, m_2; n_1, n_2) z_1^{m_1} z_2^{m_2} w_1^{n_1} w_2^{n_2}.$$

This is the “mess”: written as-is it has unbounded degree. The chain observable, however, picks out only the part of $\hat{V}$ matching the boundary occupation: *total input degree* ≤ 1 , *total output degree* ≤ 1 . This bidegree projection is the AC step that turns $\hat{V}$ into a finite object — the same coefficient-extraction move that turns $\mathcal{Z}(g, r; J)$ into the Wick weights $4 \cdot 3$ and $\binom{4}{2}^2 \cdot 2 \cdot 4! / 2$ in your §1.

Project to bidegree (1,1). Walk the four terms (each is a Kronecker-delta condition on the indices):

- T1: $m_1=n_1+2$, so input degree ≥ 2 . *Killed.*
- T2: $m_1=n_1$, $m_2=n_2$, weight n_1 . At bidegree (1,1) only (1,0;1,0) contributes (weight 1): $+z_1w_1$.
- T3: $m_2=n_2+1$ and $m_1=n_1+2$; bidegree (1,1) would force $n_2 \leq -1$. *Killed.*
- T4: $m_1=n_1+1$, $m_2=n_2-1$, weight $-n_2$. At bidegree (1,1) only (1,0;0,1) contributes (weight -1): $-z_1w_2$.

So $\widehat{V}|_{(1,1)}(z,w) = z_1w_1 - z_1w_2$. The infinitely many other terms of $\widehat{V}$ exist; they contribute zero to this observable. Add the rail-flip $\widetilde{V}$ (swap $z_1 \leftrightarrow z_2$, $w_1 \leftrightarrow w_2$):

$$\boxed{\widehat{T}(z,w) = (\widehat{V} + \widetilde{V})|_{(1,1)} = (z_1 - z_2)(w_1 - w_2).}$$

Read off the eigenvalue. The factorisation $(z_1 - z_2)(w_1 - w_2)$ separates as (input antisymmetric mode) $\times$ (output antisymmetric mode); rank as a bilinear form is 1. Substituting $z_1=1$, $z_2=-1$ collapses the polynomial to $2(w_1 - w_2)$: **eigenvalue** $\lambda = 2$. $z_1=z_2=1$ gives 0: kernel. Each rung is $\widehat{T}$, so after n rungs:

$$\langle \text{boundary}_L | \widehat{T}^n | \text{boundary}_R \rangle = \lambda^n = \boxed{2^n}.$$

Direct answers to your three sub-questions.

- *Why 2×2 , not infinite?* The boundary $|(1,0)\rangle$ at both ends restricts both input and output to total occupation 1 — the bidegree-(1,1) subspace. The full $\widehat{V}$ has unbounded degree, but the observable only sees its bidegree-(1,1) slice.
- *What is $T(z)$?* It is the bidegree-(1,1) projection of $\widehat{V} + \widetilde{V}$, with no z -dependence beyond the explicit $\{z_1, z_2, w_1, w_2\}$ used as basis labels for the boundary sector. Concretely, $\widehat{T}(z,w) = (z_1 - z_2)(w_1 - w_2)$.
- *How is it calculated?* Walk the four Kronecker conditions of V at bidegree (1,1) (one paragraph, above), add the rail-flip, factor. Total work: a half-page calculation, no Mathematica.

5. Verification

`verify_2x2.py` (accompanying this letter) does both halves:

- the polynomial collapse symbolically (sympy) — gives $\widehat{T} = (z_1 - z_2)(w_1 - w_2)$ and confirms $\lambda=2$ on the antisymmetric mode;
- brute-force enumeration over $\ell_1, \dots, \ell_{2n}$ for $n = 1, \dots, 5$ — gives 2, 4, 8, 16, 32, matching your Mathematica.

`poc_brw.py` runs the algorithmic Layer-1+2 pipeline on BRW and reproduces (i) thesis Eq. (3.25)'s $s = 2$ on the branch bubble, (ii) thesis p.91–92's count of 7 distinct 3-point 1-loop integrals, (iii) the 1+2+5 Reggeon-DP topology inventory.

Q2 takes the same machinery end-to-end through the Janssen–Täuber 2005 2-loop DP exponents.